Title: HW07—Services

Name: Gabe Everett

Email: gabe.everett@icloud.com

GitHub Repo: grceverett/mechatronics_hw

YouTube Link: https://youtu.be/LDi9Pkt571Y

Lines with **bold text** are functions from a separate file

hw07_services_main.m

Shut down any residual ros servers
Initialize new ros server
DEFINE ros robot handler (r), global dictionary (keys, values), and options (optns)
RESET the simulation [**resetWorld.m**]
GET models [**getModels.m**]
CALCULATE number (n) of models
DEFINE looping rate
FOR number (n) of models
    GET model names
    GET model poses [**get_model_pose.m**]
    PRINT position and orientation

resetWorld.m

CREATE ros service client for service “/gazebo/get_world_properties”
CREATE empty request message
CALL reset service

getModels.m

CREATE ros service client for service “/gazebo/get_world_properties”
CREATE empty request message
CALL model service to receive model names

Get_model_pose.m

DEFINE robot handle (rHandle)
CREATE empty message request for model states
POPULATE message request with model names
TRY to call the server with model names
RETURN model poses